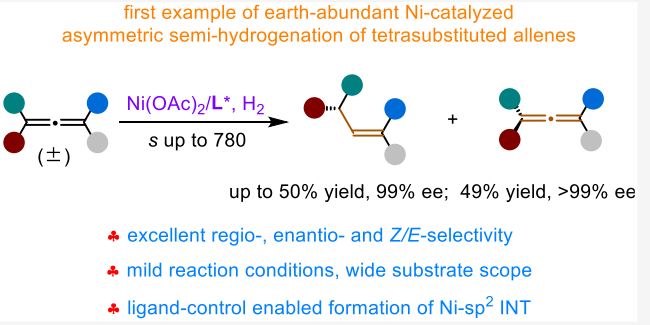
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Challenging Task of Ni-Catalyzed Highly Regio-/Enantioselective Semihydrogenation of Racemic Tetrasubstituted Allenes via a Kinetic Resolution Process

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Cite This: *J. Am. Chem. Soc.* 2024, 146, 7419−7430
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ABSTRACT: The first earth-abundant transition metal Ni-catalyzed highly regio- and enantioselective semihydrogenation of racemic tetrasubstituted allenes via a kinetic resolution process as a challenging task was well established. This protocol furnishes expedient access to a diversity of structurally important enantioenriched tetrasubstituted allenes and chiral allylic molecules with high regio-, enantio-, and *Z*/*E*-selectivity. Remarkably, this semihydrogenation proceeded with one carbon−carbon double bond of allenes, which was regioselective complementary to the Rh-catalyzed asymmetric version. Deuterium labeling experiments and density functional theory (DFT) calculations were carried out to reveal the reasonable reaction mechanism and explain the regio-/stereoselectivity.



INTRODUCTION

The chemistry of allenes with the intriguing structural architecture of the orthogonal cumulative π-system has been an area of considerable interest over the past few decades.[1−13] Allenes work as important building blocks with versatile reactivity and have always been regarded as the hybrids of alkenes and alkynes, which have been engaged in various chemical reactions to access complicated skeletons. A number of efficient transformations of mono- and disubstituted allenes are well developed in modern organic synthesis; however, tri- and tetrasubstituted allenes working as substrates have not always been an easy endeavor and are less developed (Scheme 1a).[14,15] Among them, the carbometalation of allenes with organometallic M−Nu intermediates always resulted in the chemo- and regioselective generation of two disparate carbometalated intermediates M-alkenyl INT (sp^2-M bond formation) or M-π-allyl INT (sp^3-M bond formation; Scheme 1b).[16−21] These highly reactive intermediates went through a variety of synthetic transformations and always required rare and expensive noble-metal catalysts, including protonation, annulation, β-hydride elimination, and reductive elimination, providing structurally important organic molecules. Comparatively, the transformations of thermodynamically disfavored M-alkenyl INT were challenging and reported scarcely. Among these metal-catalyzed transformations, there are limited examples involving the selective reduction of allenes, which always result in intricate reaction systems, including full hydrogenation products, four different stereoisomeric alkenes from regio- and stereoselective semihydrogenation.[22−25] Therefore, it is an enormous challenge to realize the selective reduction of allenes with high regio- and stereoselectivity. In 2017-, Dong and co-workers reported an elegant work that the highly efficient Rh-catalyzed regio- and enantioselective semireduction of 1,1-disubstituted allenes was first successfully developed in the presence of a Hantzsch ester reductant (Scheme 1c—left).[26] The more challenging tri- and tetrasubstituted allenes are difficult to achieve semireduction due to their intrinsic difficulty. Recently, Zhang, Lv, and co-workers developed Rh-catalyzed highly regio- and enantioselective semihydrogenation of tetrasubstituted allenyl sulfones to afford chiral allylic sulfones with excellent 1,2-regioselectivities and high enantioselectivities (Scheme 1c—right).[27] Considering the abundance in the earth, low price, and low toxicity of nickel,[28−34] as well as the potential to give promising reactivity and selectivity, until now, the semireduction of allenes, especially challenging and fascinating tetrasubstituted allenes substrates, promoted by cheap nickel catalysts has never been explored. Therefore, it is highly desirable and important to develop nickel-catalyzed asymmetric semihydrogenation of challenging tetrasubstituted allenes in an excellent regio/enantioselective manner, and there are several intractable

Received: November 13, 2023
Revised: February 10, 2024
Accepted: February 12, 2024
Published: March 6, 2024



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https://doi.org/10.1021/jacs.3c12597
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Scheme 1. Allenes as Transformation Precursors and Enantioselective Semireduction of Allenes

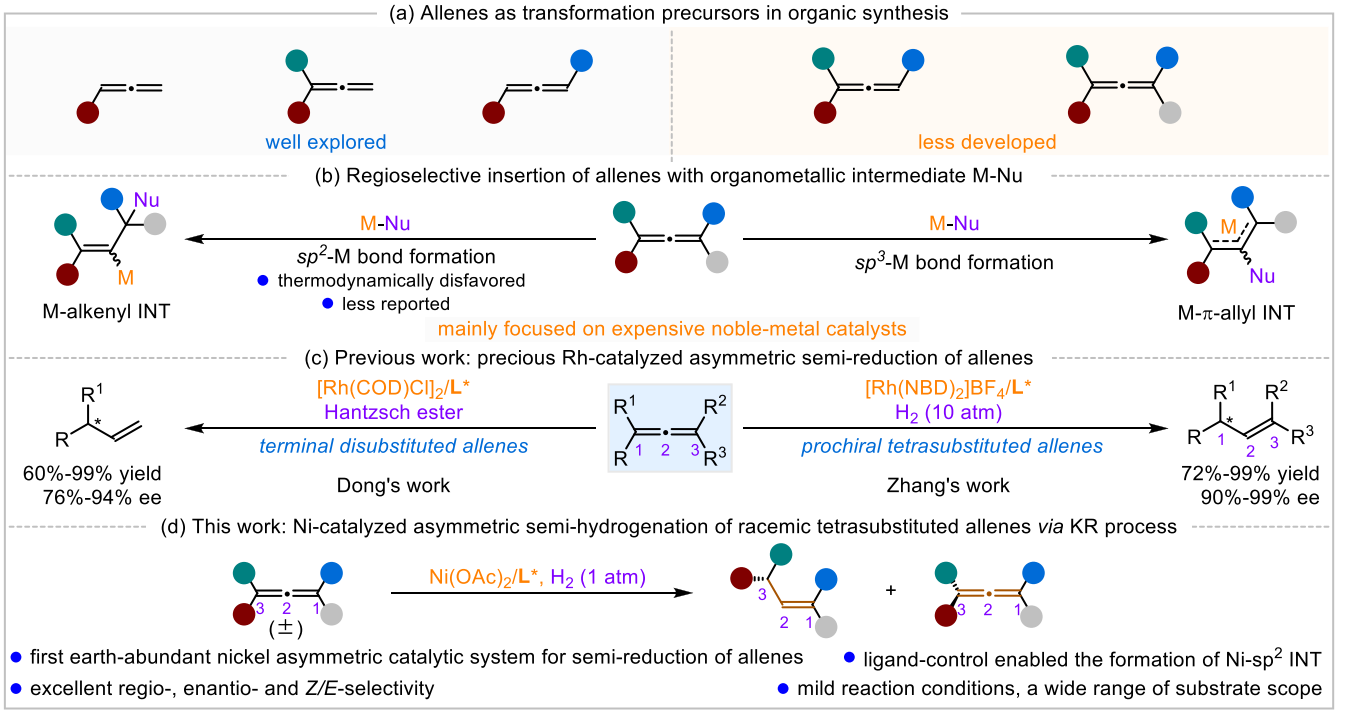
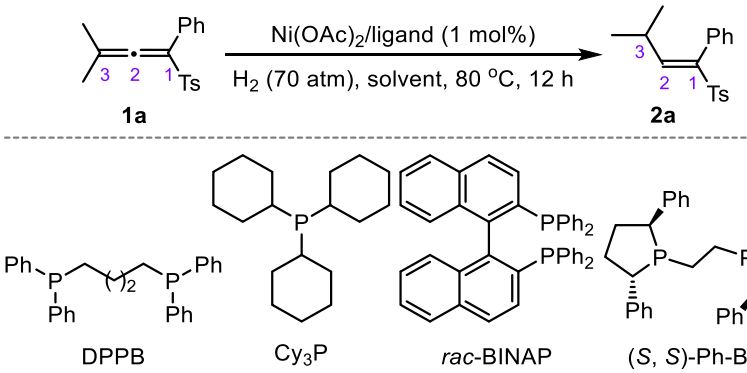


Table 1. Effect of Reaction Parameters[a]



entry	ligand	solvent	conv. (%)[b]
1	Cy_3P	HFIP	NR
2	DPPB	HFIP	NR
3	*rac*-BINAP	HFIP	NR
4	(*S*,*S*)-Ph-BPE	HFIP	63
5	(*S*,*S*)-Ph-BPE	MeOH	22
6	(*S*,*S*)-Ph-BPE	^{i}PrOH	5
7	(*S*,*S*)-Ph-BPE	DCM	46
8	(*S*,*S*)-Ph-BPE	toluene	18
9[c]	(*S*,*S*)-Ph-BPE	HFIP	75
10[d]	(*S*,*S*)-Ph-BPE	HFIP	>99

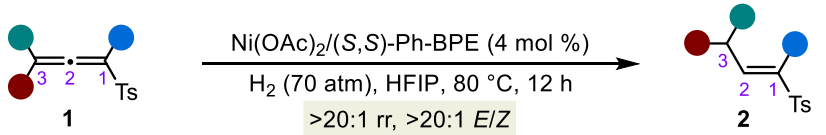
[a]All reactions were carried out with 0.1 mmol of **1a** in 1.0 mL of solvent in the presence of 1 mol % $Ni(OAc)_2$/ligand. The catalyst was pre-complexed in HFIP and MeOH. [b]Conversion was determined by crude ^{1}H NMR analysis. [c]2 mol % $Ni(OAc)_2$/(*S*,*S*)-Ph-BPE. [d]4 mol % $Ni(OAc)_2$/(*S*,*S*)-Ph-BPE.

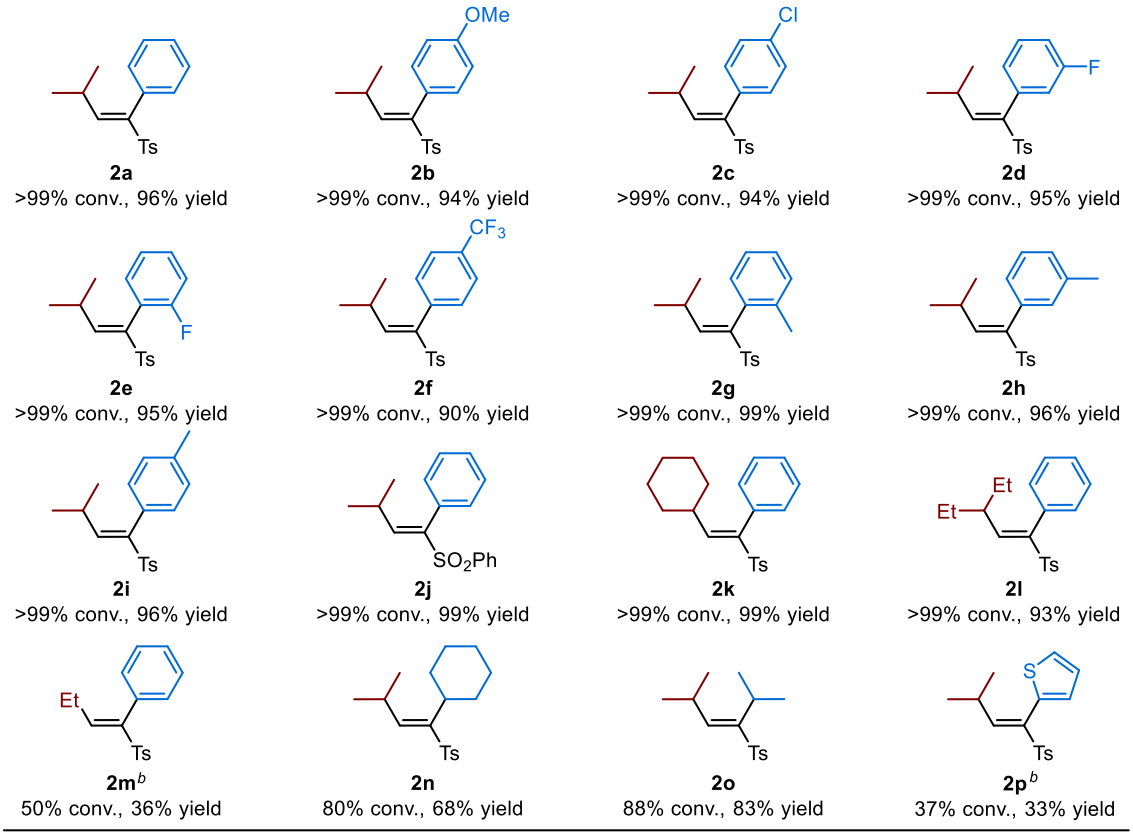
issues to overcome, namely, (1) the semihydrogenation of allenes promoted by an earth-abundant metal catalyst remains unknown, (2) how to control the regio-, enantio-, and *Z*/*E*-selectivity of semihydrogenation of allenes, and (3) the cyclization or polymerization of allenes may result in low reaction efficiency and poor selectivity.

On the other hand, considering the intriguing peculiarities of these two cumulated C═C bonds and the special axial chirality, the optically active allenes are not only abundant in plenty of natural products, biologically active compounds, and functional materials but also proven as versatile synthetic precursors for the preparation of other important mole-

https://doi.org/10.1021/jacs.3c12597
J. Am. Chem. Soc. 2024, 146, 7419−7430

Table 2. Substrate Scope Study for Ni-Catalyzed Nonenantioselective Semihydrogenation of Tetrasubstituted Allenes[a]





[a]All reactions were carried out with 0.1 mmol of **1a** in 1.0 mL of solvent in the presence of 4 mol % $Ni(OAc)_2$/(*S*,*S*)-Ph-BPE. Conversion was determined by crude ^1H NMR analysis. The yield was isolated yield. [b]10 mol % $Ni(OAc)_2$/(*S*,*S*)-Ph-BPE, 70 atm, 12 h.

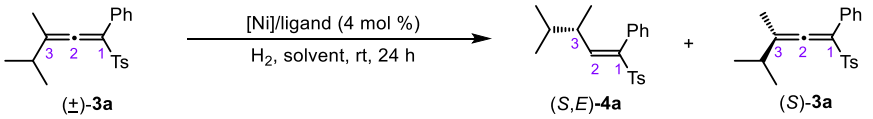
cules.[35−42] In addition, tetrasubstituted axially chiral allenes worked as privileged chiral ligands and organocatalysts in asymmetric catalytic synthesis.[43,44] As a consequence, great efforts have been made toward the development of asymmetric catalytic tactics involving a single or two electron transfer process[45−49] to access these valuable axially enantioenriched allenes. At present, some catalytic enantioselective synthesis processes have been developed, such as asymmetric 1,*n*-addition to conjugated enynes and derivatives,[50−57] asymmetric cascade reactions,[58−61] asymmetric alleno-Mannich-type reaction,[62] enantioselective radical reactions with alkynes,[63,64] asymmetric multicomponent reaction,[65] organocatalytic asymmetric C(sp^2)−H allylic alkylation of racemic trisubstituted allenoates,[66] Pd-catalyzed decarboxylative *γ*-arylation of acyclic *β*,*γ*-alkynoic acids,[67] and asymmetric transformations with propargylic alcohols.[68−71] Despite these noticeable recent advances, the direct asymmetric transformations of racemic tetrasubstituted allenes to construct axially chiral tetrasubstituted allenes are still not well-established and are of great interest. Considering the continuous interest in earth-abundant nickel catalytic systems[72−81] for the hydrogenation of allenes and the importance of axially chiral tetrasubstituted allenes, we envision that Ni-catalyzed asymmetric semihydrogenation of racemic challenging tetrasubstituted allenes via a kinetic resolution process could offer the opportunity to access these two important kinds of enantiomerically enriched molecules in the meantime (Scheme 1d). Herein, we made notable efforts to develop an inexpensive, earth-abundant first-row 3d transition metal nickel catalytic system for the asymmetric semihydrogenation of tetrasubstituted allenes and present our first results on Ni-catalyzed highly regio- and enantioselective semihydrogenation of racemic tetrasubstituted allenes via a kinetic resolution process. It furnished an expedient approach to chiral allylic molecules with high regio-, enantio-, and *Z*/*E*-selectivity and axially enantioenriched tetrasubstituted allenes.

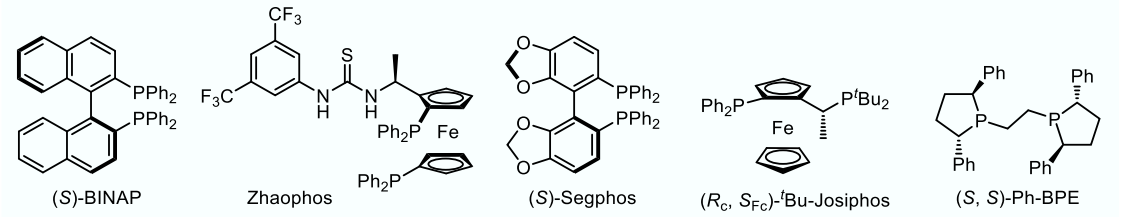
RESULTS AND DISCUSSION

Development of Nonenantioselective Ni-Catalyzed Semihydrogenation of Tetrasubstituted Allenes. To probe the feasibility of our proposal, we commenced our study by the investigation of the nonenantioselective Ni-catalyzed hydrogenation of tetrasubstituted allenyl sulfones using 1-methyl-4-((3-methyl-1-phenylbuta-1,2-dien-1-yl)-sulfonyl)benzene **1a** as the model substrate (Table 1). A series of commercially available phosphine ligands were applied to this Ni-catalyzed hydrogenation. It was found that Cy_3P, 1,4-bis(diphenylphosphino)butane (DPPB), and *rac*-

https://doi.org/10.1021/jacs.3c12597
J. Am. Chem. Soc. 2024, 146, 7419−7430

Table 3. Optimization Reaction Conditions for Ni-Catalyzed Asymmetric Semihydrogenation of Racemic Tetrasubstituted Allene 3a via a Kinetic Resolution Process[a]





entry	ligand	[Ni]	solvent	H_2 (atm)	conv. (%)[b]	ee of **4a** (%)[c]	ee of (*S*)-**3a** (%)[c]	*s*[d]
1	(*S*)-BINAP	$Ni(OAc)_2$	HFIP	70	12	38	5	2
2	Zhaophos	$Ni(OAc)_2$	HFIP	70	6	23	0	NA
3	(*S*)-Segphos	$Ni(OAc)_2$	HFIP	70	7	12	0	NA
4	(R_C,S_{FC})-Josiphos	$Ni(OAc)_2$	HFIP	70	NR	NA	NA	NA
5	(*S*,*S*)-Ph-BPE	$Ni(OAc)_2$	HFIP	70	64	61	>99	20
6	(*S*,*S*)-Ph-BPE	$Ni(OAc)_2$	TFE	70	59	60	>99	20
7	(*S*,*S*)-Ph-BPE	$Ni(OAc)_2$	CH_2Cl_2	70	54	84	>99	61
8	(*S*,*S*)-Ph-BPE	$Ni(OAc)_2$	THF	70	NR	NA	NA	NA
9	(*S*,*S*)-Ph-BPE	$Ni(OAc)_2$	dioxane	70	NR	NA	NA	NA
10	(*S*,*S*)-Ph-BPE	$Ni(OAc)_2$	hexane	70	64	55	>99	17
11	(*S*,*S*)-Ph-BPE	$Ni(OAc)_2$	$CHCl_3$	70	14	93	9	4
12	(*S*,*S*)-Ph-BPE	$Ni(OAc)_2$	CH_2Cl_2	10	54	84	>99	61
13[e]	(*S*,*S*)-Ph-BPE	$Ni(OAc)_2$	CH_2Cl_2	1	50	94	>99	170
14[e]	(*S*,*S*)-Ph-BPE	$Ni(OTf)_2$	CH_2Cl_2	1	7	98	8	217
15[e]	(*S*,*S*)-Ph-BPE	$Ni(DME)Cl_2$	CH_2Cl_2	1	9	98	10	215
16[e]	(*S*,*S*)-Ph-BPE	$Ni(ClO_4)_2$	CH_2Cl_2	1	20	98	24	242
17[e]	(*S*,*S*)-Ph-BPE	$Ni(acac)_2$	CH_2Cl_2	1	NR	NA	NA	NA

[a]Unless otherwise noted, all reactions were carried out with 0.1 mmol of (±)-**3a** in 1.0 mL of solvent in the presence of 4 mol % $Ni(OAc)_2$/(*S*,*S*)-Ph-BPE. The catalyst was pre-complexed in HFIP and MeOH. NR is no reaction, and NA is not available. [b]Conversion was determined by crude ^{1}H NMR analysis. [c]ee value was determined by chiral HPLC analysis. [d]$s = \ln[(1 - \text{conv})(1 - \text{ee}_1)]/\ln[(1 - \text{conv})(1+ \text{ee}_1)]$, $\text{conv} = \text{ee}_1/(\text{ee}_1 + \text{ee}_2)$. [e]The reaction time is 12.5 h.

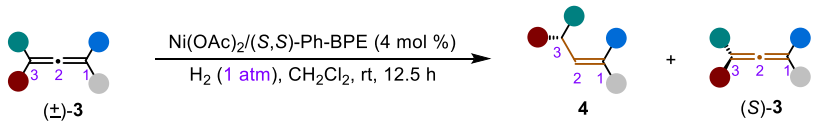
BINAP did not promote this reaction, and no reaction was detected (Table 1, entries 1−3). Fortunately, product **2a** with the semireduction at 2,3-position was obtained with moderate conversion using (*S*,*S*)-Ph-BPE as the ligand, and no other isomer byproducts were not observed (63% conversion, Table 1, entry 4). A variety of solvents were then evaluated and indicated that HFIP was still the optimal solvent. DCM could provide moderate conversion (46% conversion, Table 1, entry 7). This transformation did not perform well in MeOH, ^{i}PrOH, and toluene, with poor reactivities (5−22% conversions, Table 1, entries 5−6, 8). When the catalyst loading was gradually increased from 1 to 4 mol %, full conversion was achieved to afford product **2a** (Table 1, entries 9 and 10).

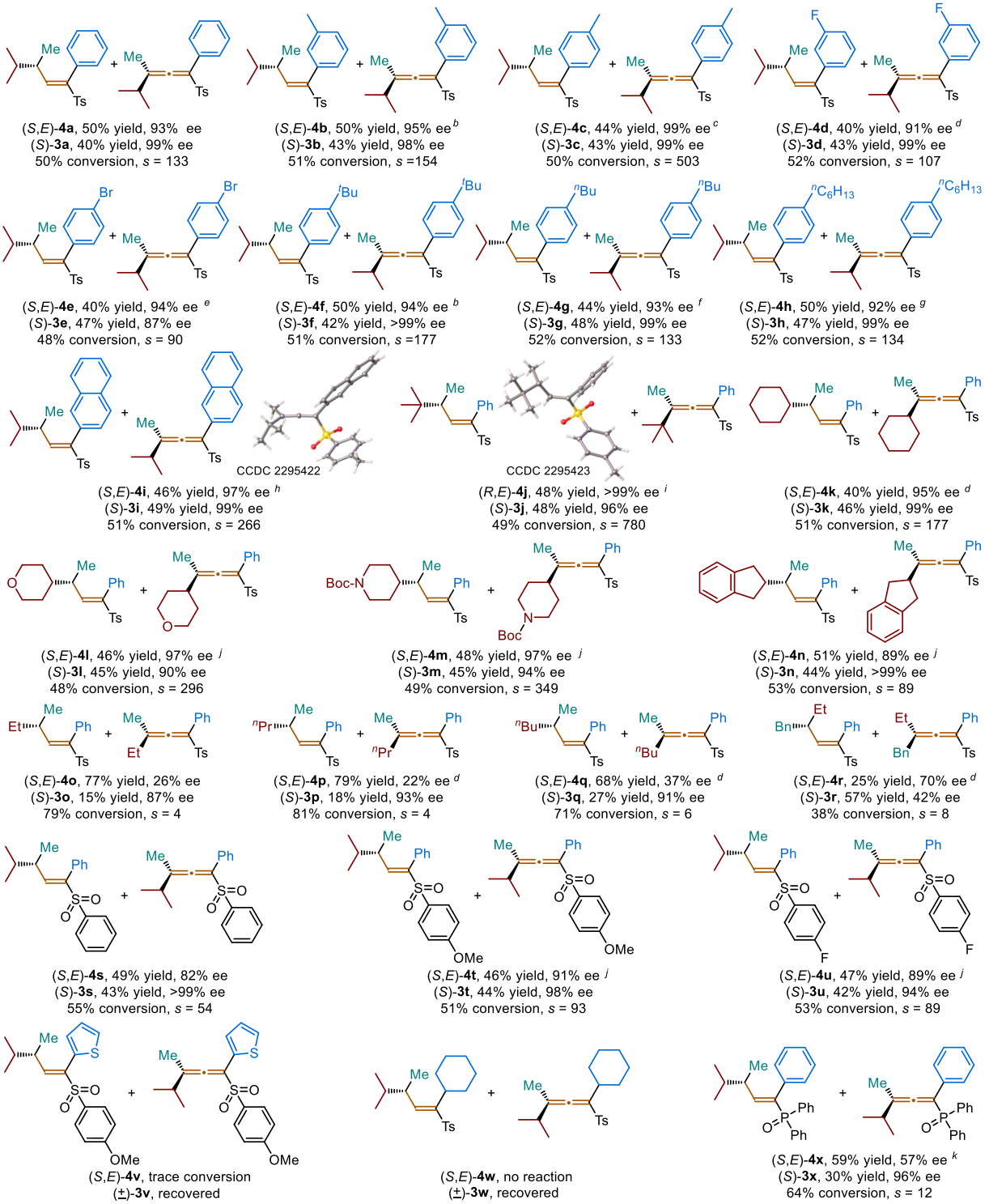
After the optimization of reaction conditions, we focused on the investigation of the substrate scope generality of this Ni-catalyzed nonenantioselective semireduction of tetrasubstituted allenyl sulfones. As summarized in Table 2, it was found that a diverse range of tetrasubstituted allenyl sulfones are viable in this transformation, generating various alkenyl sulfones in good to high yields with excellent regioselectivities. A variety of tetrasubstituted allenyl aryl sulfones with two methyl groups, bearing electron-rich (OMe, CH_3) or electron-poor (with F, Cl, or CF_3) substituents on the phenyl ring, were hydrogenated smoothly to furnish the corresponding semireduction products **2a**−**2i** with full conversions and high selectivities (full conversions, 90 → 99% yield, >20:1 rr, >20:1 *E*/*Z*). Substrate **1j** bearing a benzenesulfonyl group was also amenable to give the desired product **2j** in 99% yield. In addition, when the two methyl groups of these tetrasubstituted allenyl sulfones were switched to other alkyl groups, substrates **1k**−**1m** worked as suitable reaction partners to give expected semireduction products **2k**−**2m** with 36−99% yields and high regio/stereoselectivities. Furthermore, alkyl-substituted substrates, such as cyclohexyl (**1n**) and isopropyl (**1o**) groups, were compatible to generate the desired products (**2n**−**2o**) in moderate to good yields. To our delight, the tetrasubstituted allenyl sulfone (**1p**) containing the heteroaryl thiophene group was tolerated to afford the expected product (**2p**) with moderate reactivity and *E*/*Z*-selectivity (3:1).

Development of Asymmetric Ni-Catalyzed Semihydrogenation of Challenging Tetrasubstituted Allenes. Encouraged by the success of Ni-catalyzed nonenantioselective semireduction of tetrasubstituted allenyl sulfones, we are interested in developing the asymmetric version of this protocol. We envisioned that such an asymmetric semihydrogenation reaction could be realized using racemic tetrasubstituted allene substrates, and the potential kinetic resolution (KR) process may offer the

https://doi.org/10.1021/jacs.3c12597
J. Am. Chem. Soc. 2024, 146, 7419−7430

Table 4. Substrate Scope Study for Ni-Catalyzed Asymmetric Semihydrogenation of Tetrasubstituted Allenyl Sulfones[a]

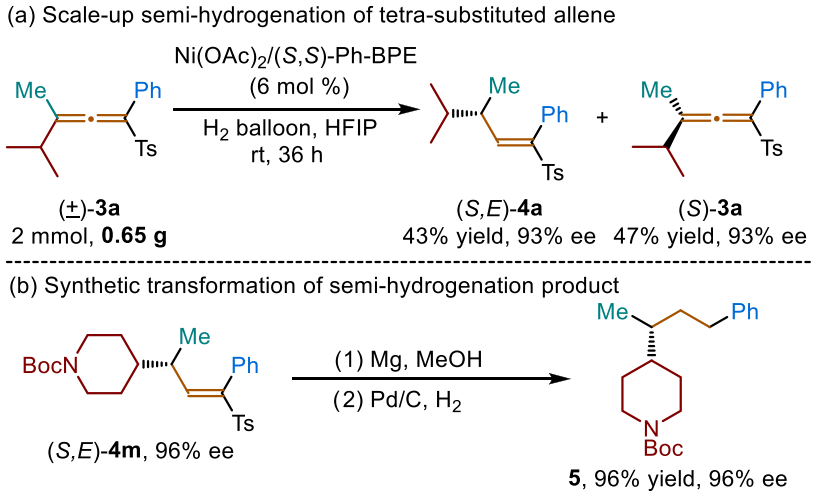




[a]Unless otherwise noted, all reactions were carried out with 0.1 mmol of (±)-**1a** in 1.0 mL of solvent in the presence of 4 mol % $Ni(OAc)_2$/(*S*,*S*)-Ph-BPE. The conversion was determined by crude ^1H NMR analysis. The ee value was determined by chiral HPLC analysis. $s = \ln[(1 - conv)(1 - ee_1)]/\ln[(1 - conv)(1 + ee_1)]$, $conv = ee_1/(ee_1 + ee_2)$. [b]10 atm, 30 °C, 36 h. [c]10 mol % $Ni(OAc)_2$/(*S*,*S*)-Ph-BPE, 70 atm, 60 °C, 20 h. [d]13 h. [e]8 mol % $Ni(OAc)_2$/(*S*,*S*)-Ph-BPE, 40 atm, rt, 50 h. [f]40 atm, 30 °C, 12.5 h. [g]8 mol % $Ni(OAc)_2$/(*S*,*S*)-Ph-BPE, 70 atm, 70 °C, 24 h. [h]10 mol % $Ni(OAc)_2$/(*S*,*S*)-Ph-BPE, 70 atm, 70 °C, 72 h. [i]30 °C, 36 h. [j]70 atm, 70 °C, 40 h. [k]30 atm, HFIP, 25 °C, 40 h.

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J. Am. Chem. Soc. 2024, 146, 7419−7430

Scheme 2. Scale-Up Synthesis and Synthetic Transformation of the Semihydrogenation Product

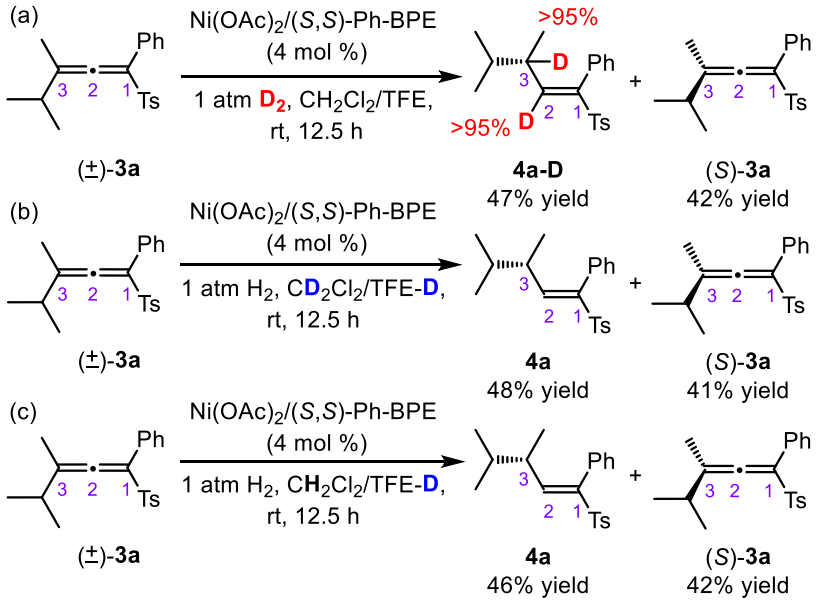


opportunity to access both chiral tetrasubstituted allenes and enantioenriched alkenyl molecules. Therefore, the initial study began with racemic tetrasubstituted allenyl sulfone **3a** as the model substrate to optimize the reaction conditions. As shown in Table 3, a broad array of chiral diphosphine ligands were utilized for this Ni-catalyzed asymmetric transformation. (*S*)-BINAP, ZhaoPhos, and (*S*)-Segphos could carry out this semireduction through the KR process but resulted in the desired product **4a** and recovered **3a** with poor reactivities and enantioselectivities, corresponding to a very poor *s*-factor (6−12% conversions, 12−38% ee for product (*S*,*E*)-**4a**, Table 3, entries 1−3). Notably, this semihydrogenation could proceed with a relative electron-rich C═C double bond of allenes, which is quite different from the previous rhodium catalytic system.[27] (R_C,S_{FC})-Josiphos did not work well in this transformation, and no reaction was observed (Table 3, entry 4). Encouragingly, (*S*,*S*)-Ph-BPE provided dramatically improved results with an *s*-factor of 20 (64% conversion, 61% ee for (*S*,*E*)-**4a**, >99% ee for the recovered (*S*)-**3a**, entry 5). Subsequently, the investigation of reaction solvents indicated that the solvent had a significant influence on the reactivity and stereoselectivity, which suggested that CF_3CH_2OH, CH_2Cl_2, and hexane could give good KR performance, the semireduction product (*S*,*E*)-**4a** could be accessible in 54−64% conversions with 55−84% ee, and the recovered (*S*)-**3a** was obtained with >99% ee (Table 3, entries 6, 7, and 10). Therefore, CH_2Cl_2 was proven to be the best solvent choice in this transformation (Table 3, entry 7). When the hydrogen pressure was reduced from 70 to 10 atm, almost the same results were generated (Table 3, entry 12 vs entry 7). To our delight, excellent results were afforded with good KR performance under very mild reaction conditions with 1 atm H_2 (50% conversion, 94% ee for (*S*,*E*)-**4a**, >99% ee for the recovered (*S*)-**3a**, *s*-factor = 170, entry 13). In addition, other nickel metal precursors, such as $Ni(OTf)_2$, $Ni(DME)Cl_2$, $Ni(ClO_4)_2$, and $Ni(acac)_2$, were then examined in this transformation under these mild reaction conditions, and no better results were observed in this semihydrogenation (Table 3, entries 14−17).

With the mild reaction conditions established, we then sought to explore the substrate scope of this Ni-catalyzed asymmetric semihydrogenation of tetrasubstituted allenes. These reaction results are listed in Table 4. A series of racemic aryl-substituted tetrasubstituted allenyl sulfones bearing methyl and isopropyl groups were first exploited; a wide range of substituted phenyl groups, which were electron-neutral (**3a**), electron-donating (**3b−c**, **3f−h**), and electron-withdrawing (**3d−e**), could be well tolerated with excellent reactivities, enantioselectivities, and KR performance (*s*-factor up to 503). It was found that the length of the alkyl-substituted group on the phenyl ring, such as *t*-butyl (**3f**), *n*-butyl (**3g**), and *n*-hexyl (**3h**) groups, did not affect the reaction results in this asymmetric transformation. To our delight, the 2-naphthyl group with steric hindrance (**3i**) was also tractable in this protocol, leading to the desired semihydrogenation product (*S*,*E*)-**4i** and recovered (*S*)-**3i** with excellent results (*s*-factor is 266). When the isopropyl group of the substrate was replaced by other sterically hindered alkyl substituents, for example, *tert*-butyl (**3j**), cyclohexyl (**3k**), tetrahydro-2H-pyran-4-yl (**3l**), *tert*-butyl piperidine-1-carboxylate-4-yl (**3m**), and 2,3-dihydro-1H-indene-2-yl (**3n**) groups, they could still proceed smoothly to give the corresponding products via an efficient KR process with excellent results. We next explored the tetrasubstituted allenyl sulfone substrates bearing other linear alkyl groups, such as ethyl (**3o**), *n*-propyl (**3p**), and *n*-butyl (**3q**) groups, which were compatible to give the recovered allenyl sulfones with excellent ee values, albeit giving semihydrogenation products (**4o−4q**) with diminished enantioselectivities. It is possible that there is no discernible difference between these linear alkyl groups and the methyl group, which led to the remarkable difficulty of chiral catalysts in providing sufficient enantiofacial differentiation during the reaction process. Moreover, the tetrasubstituted allenyl sulfone substrate (**3r**) bearing an ethyl group and a benzyl group worked well with moderate results and a diminished *s*-factor of 8. We also investigated the effect of the sulfonyl group on the reactivity and selectivity. A range of tetrasubstituted allenyl sulfone substrates bearing different electron-property groups on the phenyl ring of the sulfonyl group (**3s−3u**) were applied in this transformation. The expected semihydrogenation products (**4s−4u**) and recovered (*S*)-**3s−3u** can be available with moderate to excellent results (46−49% yields and 82−91% ee for semireduction products **4s−4u**; 42−44% yields and 94 → 99% ee for the recovered products **3s−3u**, *s*-factor up to 93). The challenging tetrasubstituted allenes containing a hetero-aromatic thienyl ring (**3v**) or three alkyl substituents (**3w**)

https://doi.org/10.1021/jacs.3c12597
J. Am. Chem. Soc. 2024, 146, 7419−7430

Scheme 3. Deuterium Labeling Experiments



were then examined in this catalytic system. Very poor conversion was observed in the semihydrogenation reaction system of thienyl-substituted tetrasubstituted allene **3v**, and no reaction was obtained using the tetrasubstituted allene containing three alkyl substituents **3w** as the substrate. Remarkably, except for the sulfonyl group, tetrasubstituted allene containing a phosphine oxide group **3x** also worked as a suitable reaction partner in this Ni-catalyzed asymmetric semihydrogenation, and the desired semihydrogenation product (**4x**) and recovered (*S*)-**3x** were accessible with moderate to excellent results (59% yield and 57% ee for semireduction product **4x**; 30% yield and 96% ee for the recovered product **3x**, *s*-factor is 12). In addition, trisubstituted allene without the Ts group 1-(3,4-dimethylpenta-1,2-dien-1-yl)-4-methoxybenzene was also examined, and a mess reaction system was observed in this transformation. The absolute configurations of the semihydrogenation products and the recovered tetrasubstituted allenyl sulfones were assigned by analogy to chiral compounds **3i** and **4j**, whose structures were unambiguously determined by X-ray crystallography analysis (CCDC 2295422 and 2295423).[82]

Scale-Up Experiment. To display the potential application of this protocol, as shown in Scheme 2a, the scale-up semihydrogenation of **3a** in 2 mmol was conducted in the presence of a H_2 balloon. To our delight, the reduction product (*S*,*E*)-**4a** and recovered **3a** could be readily accessible in excellent results without loss of enantioselectivity (for (*S*,*E*)-**4a**: 43% yield, 93% ee; for (*S*)-**3a**: 47% yield, 93% ee). In addition, the Ts group of product (*S*,*E*)-**4m** was easily removed by the treatment of Mg in MeOH and then reduced in the presence of Pd/C and H_2 to give the corresponding product **5** in 96% yield with a maintained ee value.

Reaction Mechanism Investigations. In order to gain insights into the possible reaction mechanism of this Ni-catalyzed asymmetric semihydrogenation through the KR process, deuterium labeling experiments were conducted. As shown in Scheme 3a, this asymmetric transformation of model substrate racemic **3a** was carried out under 1 atm D_2 in DCM and TFE. Semireduction product **4a-D** was obtained, and the deuterium atoms were attached at both 2- and 3-positions. Additionally, this reaction was repeated in CD_2Cl_2 and TFE-D, and product **4a** was generated without a deuterium atom. Moreover, a similar result was afforded in CH_2Cl_2 and TFE-D. As a consequence, the results of these deuterium labeling experiments demonstrated that the two deuterium atoms of semireduction product **4a-D** should be from D_2.

Density functional theory (DFT) calculations were further performed to investigate the detailed mechanism and the role of the (*S*,*S*)-Ph-BPE ligand on regio- and stereoselectivity (Figure 1).[83] Based on literature reports on phosphine-coordinated nickel complexes, dimeric nickel complex **INT1A** was set as the relative zero point for Gibbs free energy (Figure 1a).[84] Initially, the loss of one acetate ligand by intermediate **INT1A** results in the formation of dimeric nickel cation intermediate **INT2A** with an endothermal energy of 3.7 kcal·mol^{-1}. The decomposition of intermediate **INT2A** causes the generation of cationic intermediate **INT3A**, which drives the catalytic cycle of enantioselective semihydrogenation. Meanwhile, the resulting neutral intermediate **INT3B** dimerizes to regenerate **INT1A**. Subsequently, intermediate **INT3A** undergoes acetate-assisted H_2 insertion through six-membered nickel-acycle transition state **TS4A** with an activation energy of 21.2 kcal·mol^{-1} to deliver Ni−H species **INT4A** (Figure 1a). The regioselectivity of the hydrogenation can be illustrated by the frontier molecular orbital theory (Figure 1b). For the structure of allenyl sulfone, its highest occupied molecular orbital (HOMO) is closely related to both the electron-poor C2−C3 double bond and lone pair electrons of the O atom, and its HOMO−1 orbital is associated with the C1−C2 double bond. The lowest unoccupied molecular orbital (LUMO) of the Ni−H species is contributed by the Ni 3d orbital, which is 3.9 eV higher than the HOMO orbital of allenyl sulfone and 4.4 eV higher than the HOMO−1 of allenyl sulfone. The smaller LUMO−HOMO energy gap illustrates that the Ni−H species preferentially interacts with the HOMO orbital of allenyl sulfone. When the oxygen of the two enantiomers **3a** coordinates with the Ni(II) center of **INT4A** to replace acetic acid, intermediates **INT5A-S** and **INT5A-R** are generated, respectively. The following Ni−H bond is inserted into allene from the Re or Si face via transition

https://doi.org/10.1021/jacs.3c12597
J. Am. Chem. Soc. 2024, 146, 7419−7430

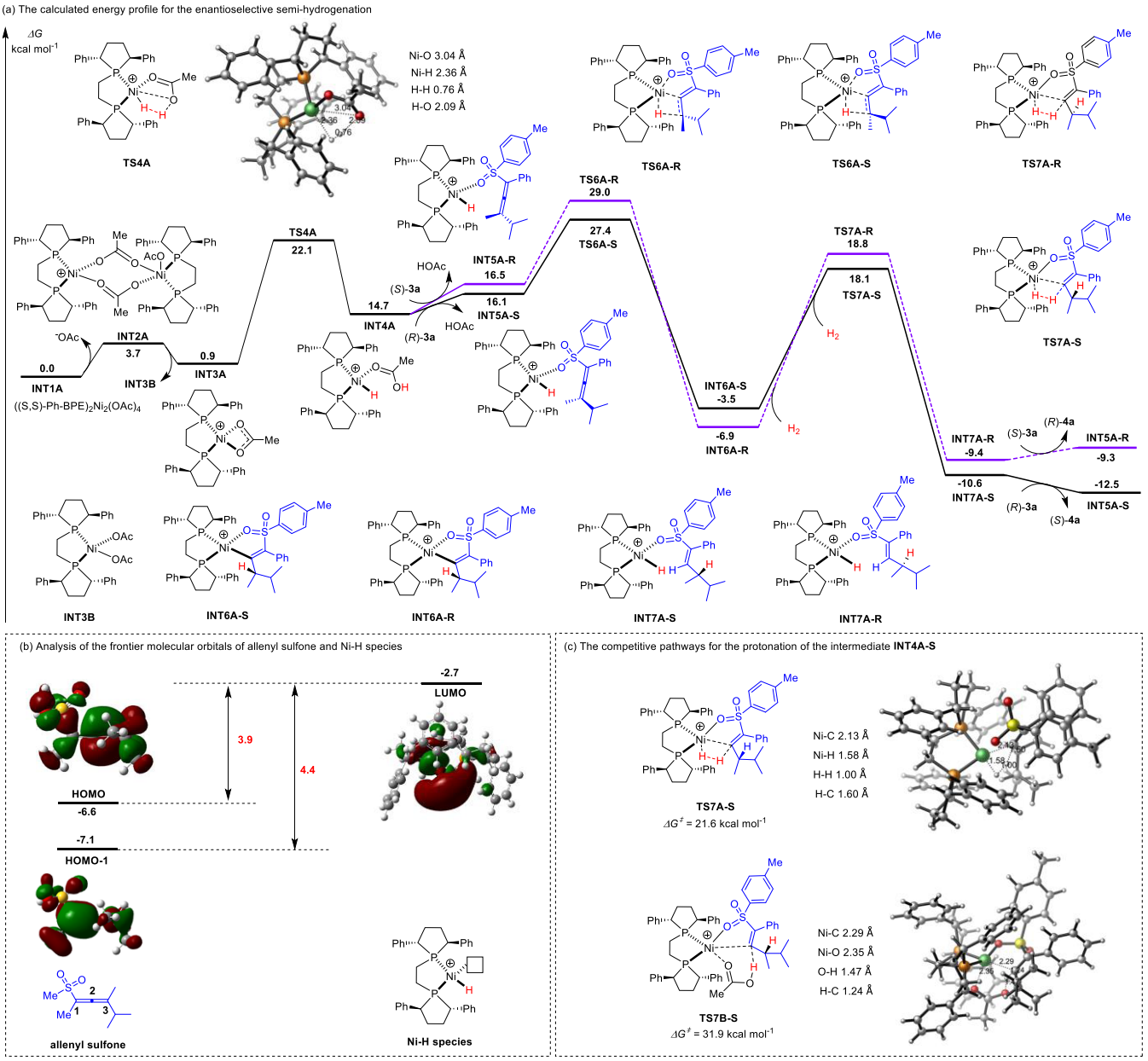


Figure 1. DFT investigations.

state **TS6A-S** or **TS6A-R**. This step is the enantio-determining step, and the pathway determining the final *S*-configuration of the hydrogenated products via **TS6A-S** is 1.6 kcal·mol^{-1} more favorable than that for the *R*-configuration pathway (12.7 kcal·mol^{-1} vs 14.3 kcal·mol^{-1} relative to **INT4A**). This computed energy difference indicates that the hydrogenation of (*R*)-**3a** is preferred, whereas the conversion of (*S*)-**3a** is thermodynamically less efficient, leading to the formation of a hydrogenated product with *S*-configuration and the enrichment of (*S*)-**3a** with high enantioselectivity.

The origin of enantioselectivity can be further elaborated by the distortion/interaction analysis shown in Figure 2.[85] In both transition states **TS6A-S** and **TS6A-R**, the distortions of the catalyst and substrate control the enantioselectivity. The distortion energies caused by the catalyst and substrate deformation in **TS6A-S** are 17.8 and 16.7 kcal·mol^{-1}, respectively, which are 12.2 kcal·mol^{-1} lower in total than that in **TS6A-R**. This indicates that there is a significant steric hindrance between the chiral ligand and the substrate in **TS6A-R**, which determines the kinetic advantage of the allene insertion step through **TS6A-S** and promotes the kinetic resolution of **3a**. Elevating the reaction temperature to 70 °C and the hydrogen pressure to 70 atm will overcome the energy barrier between **TS6A-S** and **TS6A-R** to yield the hydrogenated product with two enantiomers. Without the direct directing of sulfone, the energy barriers in **TS6B-R** and **TS6B-S** are 3.0 and 7.3 kcal·mol^{-1} higher than in **TS6A-S**, respectively, due to the weak interactions. These higher energy barriers effectively exclude the direct insertion mechanism of **INT4A** into the C2−C3 double bond. For substrates containing linear alkyl groups, such as ethyl (**3o**), the reduction of steric hindrance results in a decrease in the degree of catalyst deformation in transition state **TS6A-R-Et**. This leads to a lower energy barrier difference between **TS6A-R-Et** and **TS6A-S-Et**, which results in the observation of low *s*-factors. The subsequent protonation of intermediate **INT6A-S**

https://doi.org/10.1021/jacs.3c12597
J. Am. Chem. Soc. 2024, 146, 7419−7430

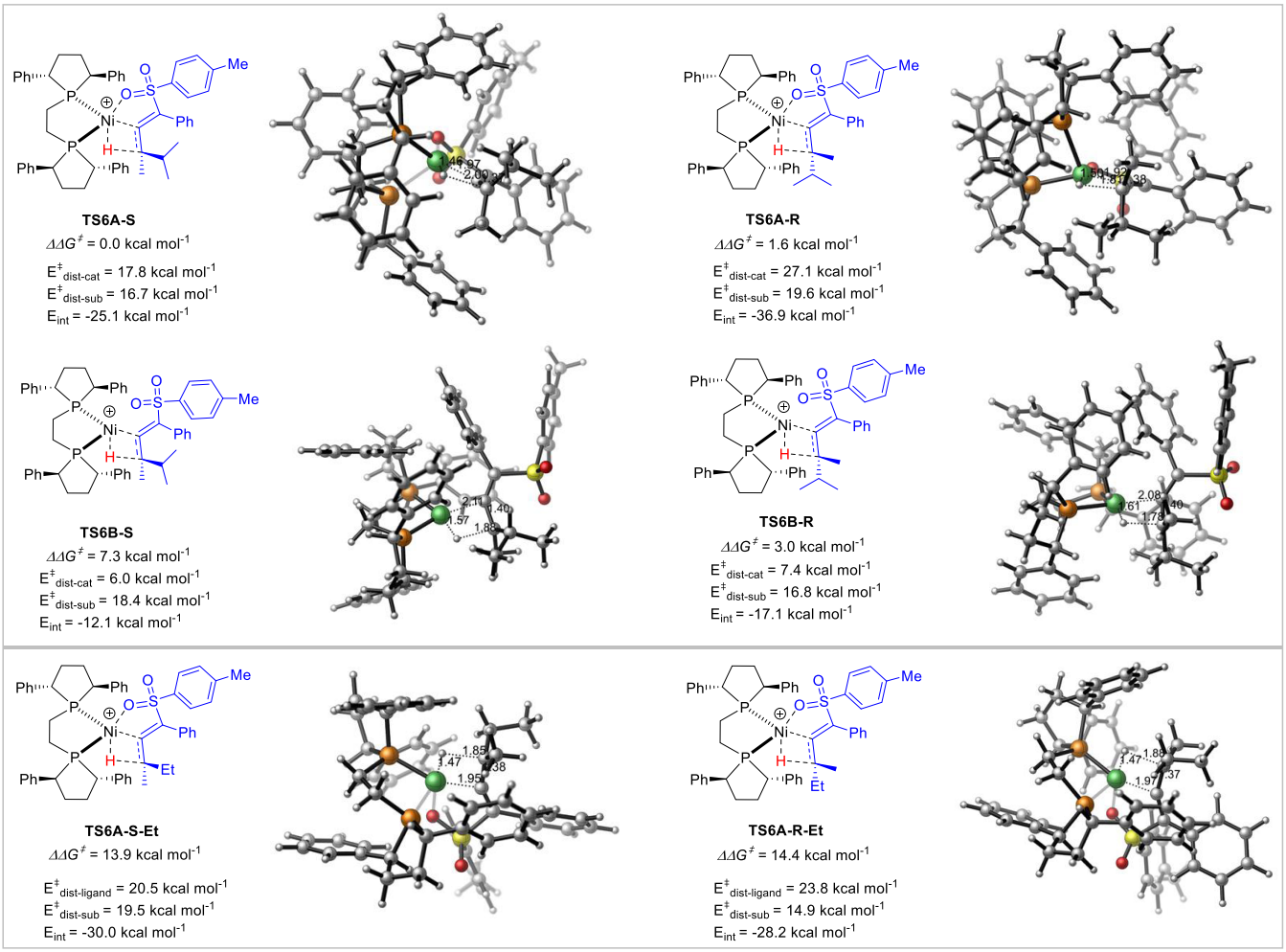


Figure 2. Stereochemical model.

occurs via two competing pathways. The more favored pathway involving the protonation with hydrogen molecule takes place through transition state **TS7A-S** with an activation energy of 21.6 kcal·mol^{-1}, much lower than the protonation by in situ-generated acetic acid through transition state **TS7B-S** (21.6 kcal·mol^{-1} vs 31.9 kcal·mol^{-1}, Figure 1c), consistent with the deuterium labeling experiments shown in Scheme 3. Finally, ligand exchange with substrate **3a** releases the desired product (*S,E*)-**4a** and regenerates active species **INT5A**.

CONCLUSIONS

In conclusion, we have realized a challenging task for the first earth-abundant transition metal Ni-catalyzed highly regio- and stereoselective semihydrogenation of racemic challenging tetrasubstituted allenes via a kinetic resolution process. Remarkably, this work presented herein nicely regioselectively complements the Rh-catalyzed asymmetric version, and one carbon−carbon double bond of allenes was hydrogenated. It represents a valuable advance in the construction of a wide range of structurally important enantioenriched tetrasubstituted allenes and chiral allylic molecules with excellent results (s-factors up to 780). Further studies of deuterium labeling experiments and DFT calculations were conducted to explain the regio-/stereoselectivity and elucidate the catalytic pathway.

ASSOCIATED CONTENT

Data Availability Statement

All data are available from the corresponding authors upon reasonable request. Crystallographic data for the structure reported in this article have been deposited at the Cambridge Crystallographic Data Center.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.3c12597.

Experimental procedures; characterization data; X-ray crystallographic analysis of chiral compounds **3i** and **4j**; NMR spectra; HPLC spectra; and DFT calculation details. (PDF)

Accession Codes

CCDC 2295422 and 2295423 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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https://doi.org/10.1021/jacs.3c12597
J. Am. Chem. Soc. 2024, 146, 7419−7430

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (22071187 and 22271226), the National Key R&D Program of China (2022YFA1503200), the Natural Science Foundation of Hubei Province (2020CFA036 and 2021CFA069), and the National Youth Talent Support Program. The authors are also grateful to the High-Performance Computing Center of Nanjing University for performing the numerical calculations in this paper on its blade cluster system.

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